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වයඹ පළාත් අධ්‍යාපන දෙපාර්තමේන්තුව
Provincial Department of Education - NWP

දෙවන වාර පරීක්ෂණය - 2025
Second Term Test - 2025

Grade 11 | **Mathematics I** | **Time : 2 hours**

Name / Index No.:

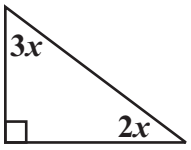
- **Answer all the questions in the paper itself.**
- **Marks will be awarded as follows ; 02 marks each for questions in part A and 10 marks each for questions in part B.**
- **(Take $\frac{22}{7}$ as the value of π)**

Part - A

01. What is the first approximation of $\sqrt{66}$. Underline the correct answer,

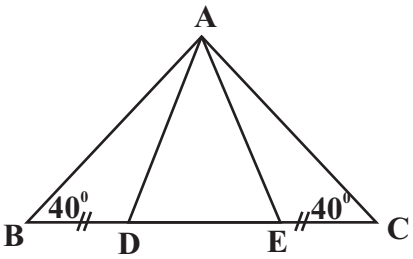
(i) **8.4** (ii) **8.3** (iii) **8.1**

02. Find the value of x according to the information given in the figure.



03. What is the smallest expression that is divisible by all three expressions $4a^2b$, $6ab^2$ and $8ab$.

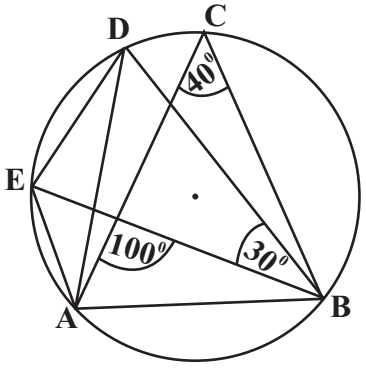
04. In the triangle **ABC**, $A\hat{B}C = A\hat{C}B = 40^0$ and **BD = EC**. Write the case which the triangles **ABD** and **AEC** are congruent.



05. Write $\log_2 \frac{1}{8} = -3$ in index form.

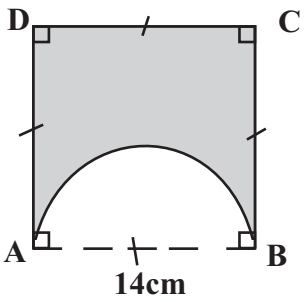
06. Simplify $\frac{2x - 4}{2x} \div \frac{x - 2}{3}$

07. **A, B, C, D, E** are the 5 points on the circle. According to the given information find the magnitude of $D\hat{A}C$.

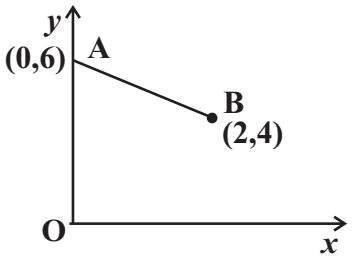


08. If an electrical appliance worth **Rs. 30 000** was imported and a custom duty of Rs. 12 000 has to be paid, find the percentage of custom duty charged.

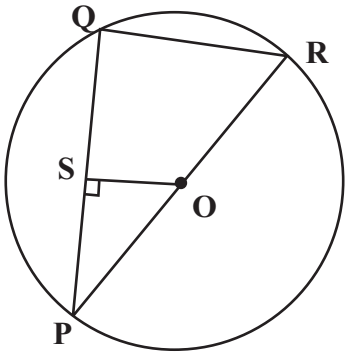
09. The remaining part after removing the semi circular section with diameter **AB** from the square **ABCD** has shaded. Find the perimeter of the shaded region.



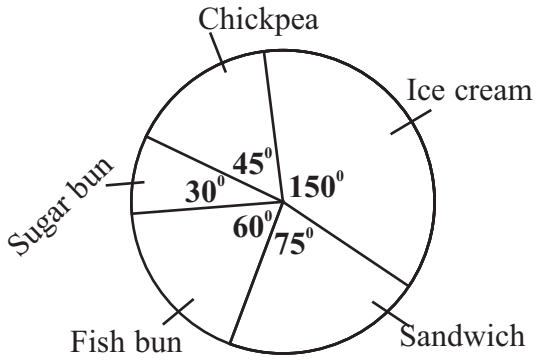
10. Write the equation of the straight line AB shown in the coordinate plane in the form of $y = mx + C$.



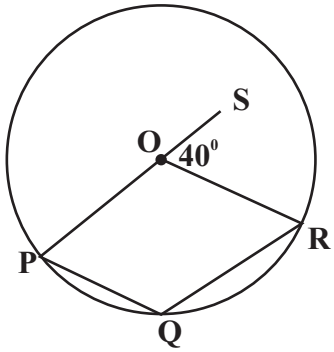
11. The center of the circle is **O**. **PQ** $\perp$ **OS**. If **PQ = 8cm** and **PR = 10cm**, find the perimeter of the quadrilateral **QROS**.



12. The pie chart shows the number of dansals that served vairous types of foods in a certain area. If 450 ice cream dansals was held, find the number of fish bun dansals were there?



13. The point **P, Q, R** are lie on the circle with center **O**. The side **PO** is extended to point **S**. If $S\hat{O}R = 40^0$, find the magnitude of $P\hat{Q}R$.

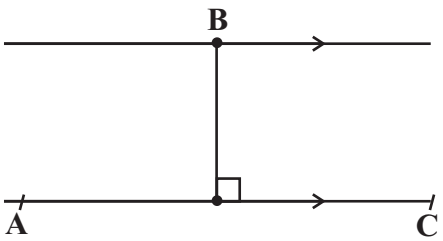


14. It takes **20** minutes to fill the empty tank if water flaws through a pipe at a rate of **80** liters per minutes. Find the capacity of the tank.

15. If $(x-3)$ is a one factor of $3x^2 - 7x - 6$, find the other factor.

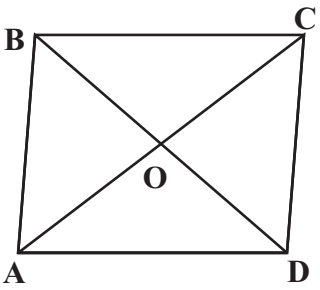
16. Write 15th term as a power of **3** of the geometric progression **3, 9, 27, ...**

17. If the angle of elevation from **A** to **B** is 50^0 and the angle of depression from **B** to **C** is 30^0 . Mark and show the information in the diagram.



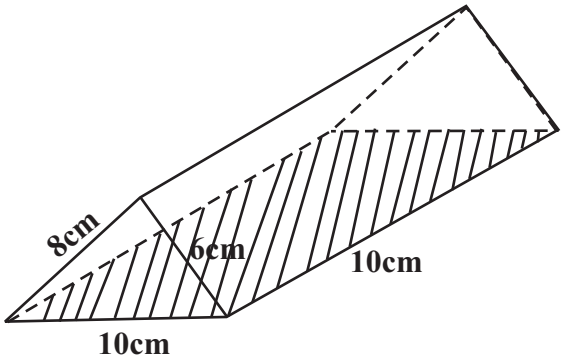
18. The sample space of a random experiment is **S = {2, 3, 4, 5, 6, 7} . A** and **B** are two sub sets. If **A = {2, 4, 6}** and **B = { 2, 3, 5, 7 }**, find $P(A \cap B)'$.

19. Based on the square **ABCD** given in the figure, the stements written by a student are shown in the tale below. put '✓' for the correct answer and '×' for the incorrect answer.

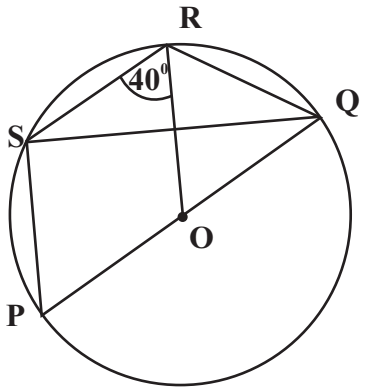


(1)	Area of the triangle **AOB** = $\frac{1}{4}$ of the area of the parallelogram **ABCD**	
(2)	**AO = OC** and **BO = OD**	
(3)	**AB = DC** and **AD = BC**	

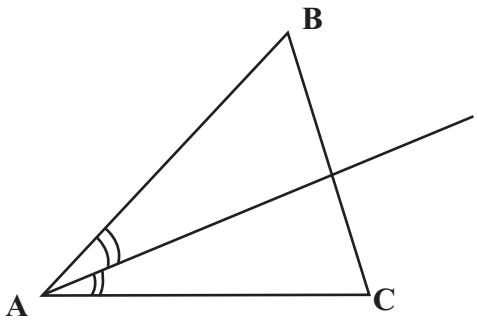
20. Draw a rough sketch with measurements of the shaded surface of the prism shown in the figure.



21. **PQRS** is a cyclic quadrilateral inscribed in a circle with center **O** and diameter **PQ.** **PORS** is a rhombus. Find the magnitude of $\hat{PQS}$.



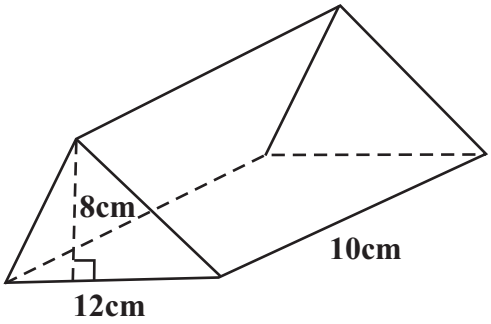
22. Three status are placed at points **A, B** and **C**. This incomplete sketch is drawn to find the approximate location of a lamp post **P** that is equidistant from **AB** and **AC** and equidistant from **A** and **B**. Complete the sketch using your knowledge of loci. Mark **P**.



23. In the table below, put '✓' mark for the correct statement and '×' mark for the incorrect statement.

(1)	The inter quartile range of data distribution **2, 3, 4, 6, 7, 8, 9** is **5.**	
(2)	The minimum of a data distribution is the lower boundary of the class interval.	
(3)	Class boundaries of the class interval **6 - 10, 11-15, 16-20** are **5.5 - 10.5, 10.5 - 15.5, 15.5 - 20.5**	

24. Find the volume of the prism shown in the figure.



25. Solve,

$$\frac{2}{x} + \frac{3}{2x} = \frac{7}{4}$$

Past Papers Wiki

Answer all the questions in the paper itself.

(01) A welfare society in a village undertakes road development using funds provided by the government for rural development. 5% from the total amount retained by the society and the remaining amount is paid to contractors for rood development. $\frac{2}{7}$ of the amount retained by the society is allocated for welfare of the members and $\frac{3}{5}$ of the remaining amount is allocated for fixed deposited.

(i) After allocating $\frac{2}{7}$ of the retained amount for welfare activity, what fraction of the amount remains?

(ii) What is the fraction of amount remains by the society After allocating funds for both welfare and fixed deposits.

(iii) If the final remaining amount with the society is Rs. 42 000, find the amount separatly allocated for welfare activities and for the fixed deposits.

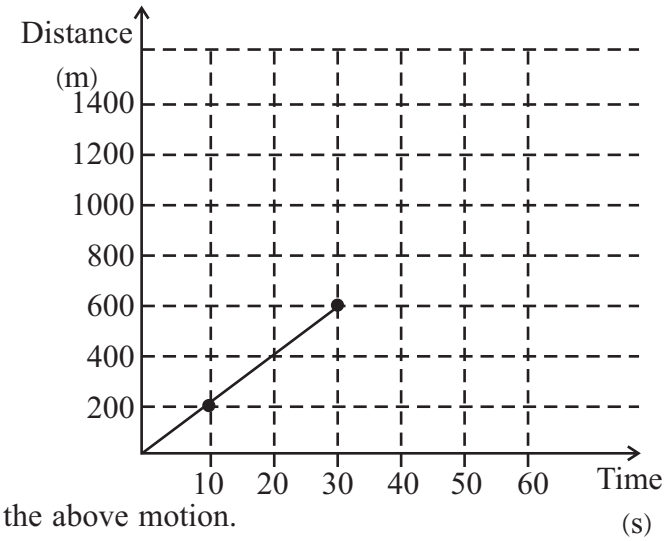
(iv) Find the total amount allocated by the government for rural developments.

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(02) (a) An incomplete table related to the distance and time of a motorcycle's motion is given and an incomplete distance - time graph has draw for it.

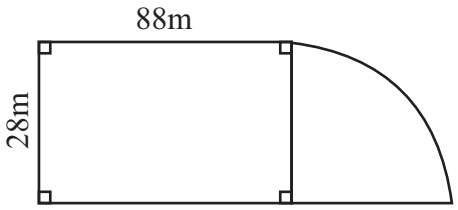
(i) Complete the table and the distance time graph.

Distance (m)	Time (s)
200	10
-	20
600	-
600	40
1000	50
1400	60



(ii) Calculate the average speed of the above motion.

(b) The given figure consists of a rectangular part with length 88m and breadth of 28m and with a sector of 90^0.



(i) Find the perimeter of composite figure.

(ii) How many times the area of the sector is the area of the composite figure.

(03) (a) To dig a canal, it is estimated that 18 people takes 10 days. In the first four days 20 people were assigned to the task.

(i) What is the total amount of work in man-days.

(ii) After first four days, how much of the work remains in man-days?

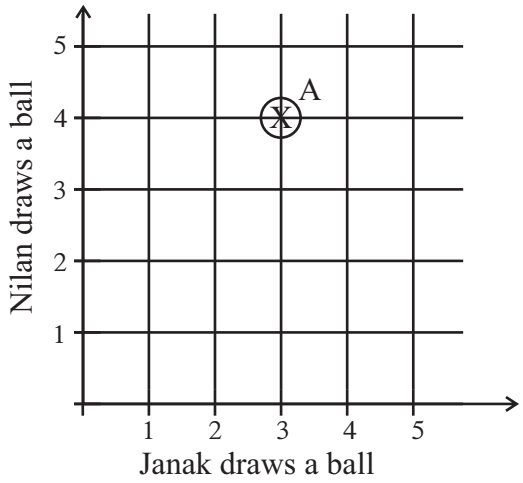
(iii) If the remaining work is to be completed in 8 days, how many additional workers need to be assigned to the task.

(b) When a camera worth Rs. 80 000 was imported, a custom duty of Rs. 23 200 has to be paid. What is the percentage of custom duty charged.

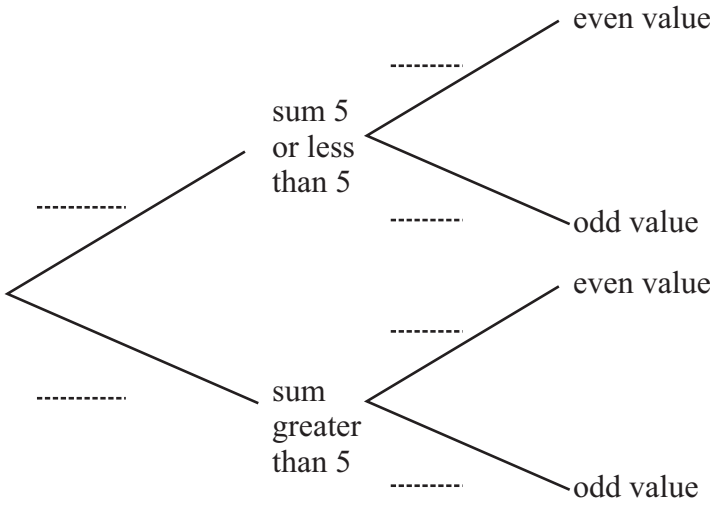
(04) (a) There are 5 balls numbered as 1, 2, 3, 4, 5 in a bag. Janaka randomly draws a ball, records its number and then returns the ball to the bag, Then Nilan draws a ball and records its number. A event of the sum of the two numbers recorded from the two draws has marked in the grid using 'X' and mark the event as A.

(i) Describe the event A.

(ii) Complete the grid using 'X" mark.



(b) (i) Complete the above tree diagram by writing probabilities that drawn based on the above random experiment.



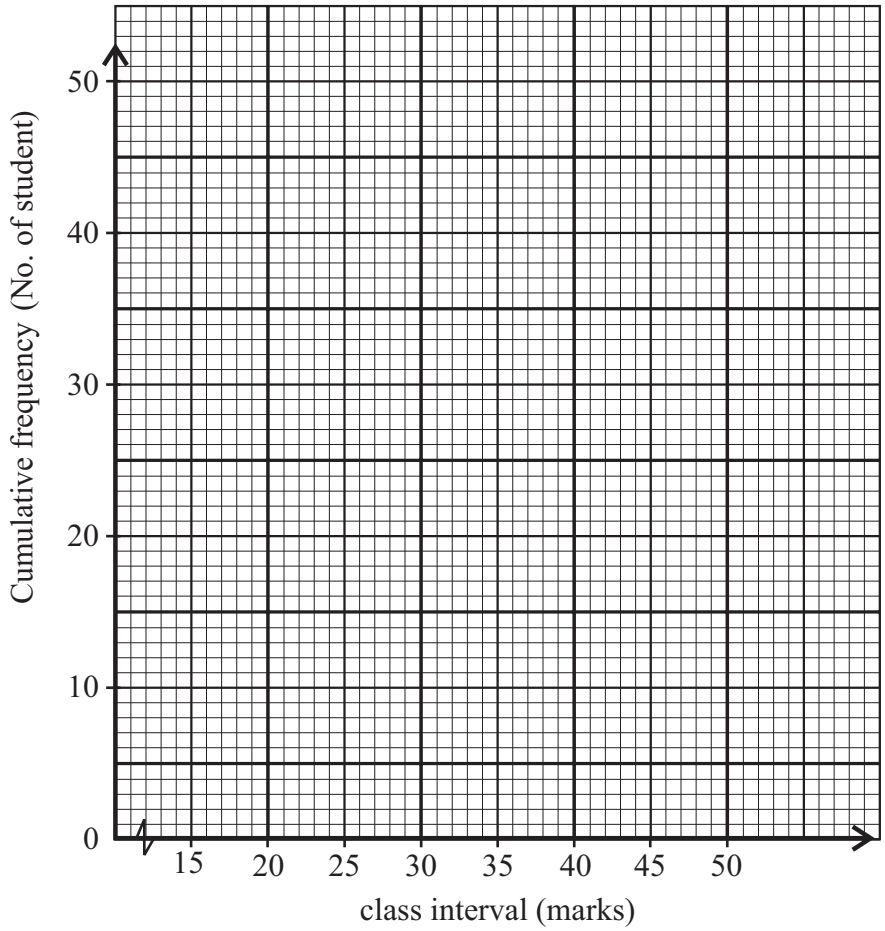
(ii) Write the probability of getting an even number greater than 5 as the sum.

(05) A frequency distribution of marks obtained by students for an assessment is given.

(i) Complete the table

Class interval (marks)	Frequency (no. of student)	Cumulative frequency
15 - 20	3	3
20 - 25	5	8
25 - 30	9	.....
30 - 35	14	31
35 - 40	8	39
40 - 45	.....	45
45 - 50	3	48

(ii) Find the median mark obtained by a student by using cumulative frequency curve.



(iii) Find the median mark obtained by a student by using cumulative frequency curve.

(iv) To select the top 25% of students, find the minimum mark a student must have to be obtained.

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Provincial Department of Education - NWP

දෙවන වාර පරීක්ෂණය - 2025 Second Term Test - 2025

Grade 11

Mathematics II

Time : 3 hours

Name / Index No.:

Additional reading time : 10 minutes. Use the additional reading time to read the question paper and select the questions and to prioritize the questions.

Instruction :

- **Select 5 questions from part A and 5 questions from part B, and answer 10 questions.**
- **When answering the questions show all relavent steps and use correct units.**
- **Each question carries 10 marks.**
- **The volume of a right pyramid with area of the base as A and height as h is $\frac{1}{3}$ Ah.**
- **The volume of sphere with radius r is $\frac{4}{3}\pi r^3$.**

PART - A

(01) Jagath borrowed a loan of Rs. 100 000 from a financial institution at an annual compound interest rate of 12%. He invested this money to buy shares from a company which pays an annual dividend of Rs. 4.50 per shares and the marked price of share was Rs. 25. In the second year, the company increased the dividend to Rs. 5 per share. At the end of the second year Jagath sold all his shares at Rs. 27 per share, and use the money to repay the loan and the interest. Show that Jagath made a profit of more than Rs. 20 500 from this transaction.

(02) An incomplete table representing corresponding y values to x values in the range $-1 \leq x \leq 5$, of the quadratic function $y = x^2 - 4x + 1$.

x	-1	0	1	2	3	4	5
y	6	.....	-2	-3	-2	1	6

(i) Find the value of y when $x = 1$ by considering the symmetry of the function.

(ii) Using the standard coordinate plane and the suitable scale, draw the graph of above quadratic function using the table in a graph paper.

(iii) (a) Write the coordinates of the two intersection points of the graph and the x - axis.

(b) Write the give quadratic function in the form of $y = (x - p)^2 - q$.(p and q are constants)

(iv) Take that the line $y = 4$ intersects the graph at the points A and B.

(a) Write the range of x when $y < 4$.

(b) Show that the length of AB is AB $\leqslant$ 5.2.

(03) (a) On a flat ground along a straight road running east - west, two points A and B are marked at a distance of 100 m apart. A radio transmission tower is visible to the north of this road such that 315^0 bearing from B and 040^0 bearing from A.

Draw a scale diagram to represents above information by representing 10 m by 1 cm. How far north of the AB road is the transmission tower located.

(b) The foot of the vertical tower PQ is P. The top of the tower is observed at an angle of elevation of 50^0 from the point R which is located 30 m away from P on the same level. Draw a scale diagram to represent above information by taking the scale 1 : 500. Find the actual height of the tower PQ. (Ignore the height of the observer)

(04) A right angled triangle with perpendicular side length $(x + 2)$ cm and x cm has cut out from the square shaped sheet with side length 9 cm. The remaining area after removing the triangle is 60 cm^2

(i) Show that x satisfy the equation $x^2 + 2x - 42 = 0$ and find the value of x in cm to the first decimal place. (Take $\sqrt{43} = 6.56$)

(ii) Show that the length of the hypothenuse of the right angled triangle is greater than 9 cm.

(05) There are two types of tiles, named type A and type B in the market. To completely cover a floor, 150 tiles of two types are needed. The price of one tile of type A is Rs. 250 and that of type B is Rs. 300. The total cost for tile is Rs. 40 800. Build up a pair of simultaneous equations by taking number of tiles in type A as x and number of tiles in type B as y. Find the number of type A tiles and type B tiles separately.

(06) The frequency distribution table below shows the number of chairs provided for a event on a rent over a month.

Class interval of no: of chairs	200 - 300	300-400	400-500	500-600	600-700	700-800	800-900	900-1000
No.: of days	1	2	4	8	6	4	3	2

(i) What is the modal class.

(ii) Find the mean number of chairs provided per day to the nearest whole number by taking the mid value of the modal class as the assumed mean.

(iii) If Rs. 5 is charged per chair, find the expected total income for the three - month period.

(iv) Show that the maximum number of chars provided during this month period exceeds 15 000.

PART - B

(07) A fence made of concrete posts is numbered from the gate. A worker repairing this fence, repairs 5 posts on the first day and then an each following days 8 more than the previous day.

(i) How many posts has the worker painted by the 7th day.

(ii) On which day will be worker complete exactly 93 posts.

(iii) If the worker finished all repairing on the 51st day, how many posts does the fence have?

(iv) If the fence numbered as 10, 20, 30, ... and both the starting and ending posts are painted orange, How many orange color painted posts, by the end of the 51st day.

(08) Use only cm/mm straightedge and a compass for the geometric constructions given below. The constructions lines must drawn clearly.

(i) Construct the triangle ABC such that AB = 7cm, $A\hat{B}C = 120^0$ and AB = BC.

(ii) Construct the perpendicular bisector of AB and mark D where it cuts the line AB.

(iii) Construct the angular bisector of $A\hat{B}C$. Mark the point E where the locus in (ii) above and the angular bisector meet each other. Mark the intersection point of AC and BE as P.

(iv) Construct the semicircle with center D that goes through the points A, B and P.

(v) Construct a parallel line to AB through P and mark the point that it cuts the semi circle as Q.

(vi) Find the magnitude of $A\hat{P}Q$.

(09) A metal right square pyramid with side length of square base is a cm and height of $3a$ cm is melted down and without any wastage, make a solid metal sphere of radius r cm.

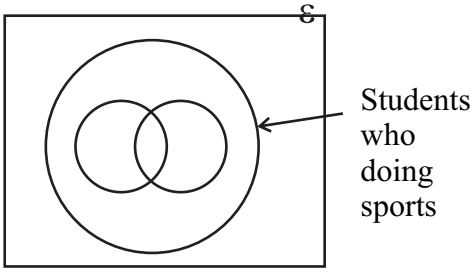
Show that, $a^3 = \frac{\pi}{9} r^3$

Find the value of a^3 using logarithmic table by taking r = 2.84 cm and π = 3.14, and find the side length of the square base to the nearest centimeter.

(10) An incomplete venn diagram drawn based on information collected from Grade 11 students are shown below.

(i) Draw the given venn diagram in your answer sheet and fill the following informations.

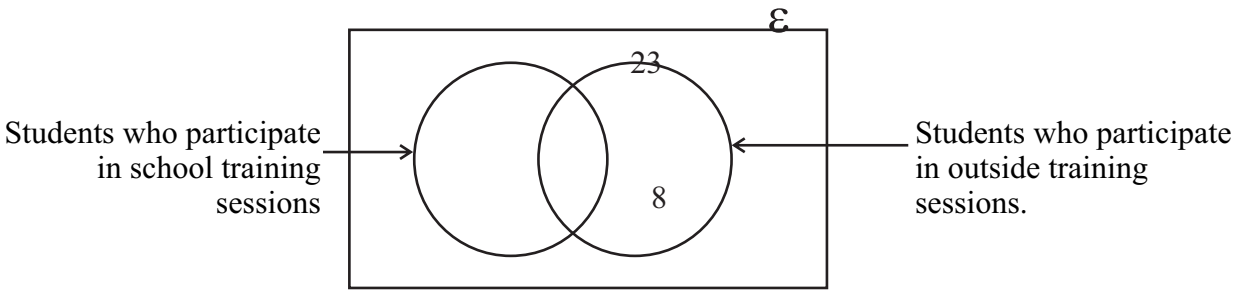
- 20 students play crickets.
- 15 students play volley ball.
- 10 students play other sports.



(ii) Find the number of students who play cricket but not volley ball.

(iii) If 9 students do not play any sport, find the total number of students.

(iv) It was found that the students who play sports also participate in training sessions both in school and out side. According, an incomplete venn diagram has drawn. Complete the venn diagram and find the probability of the students in this group participates in the school training sessions.



(11) AB is the diameter of the circle with center O. The chord CD is drawn such that CE = ED while the point E is lies on AB.

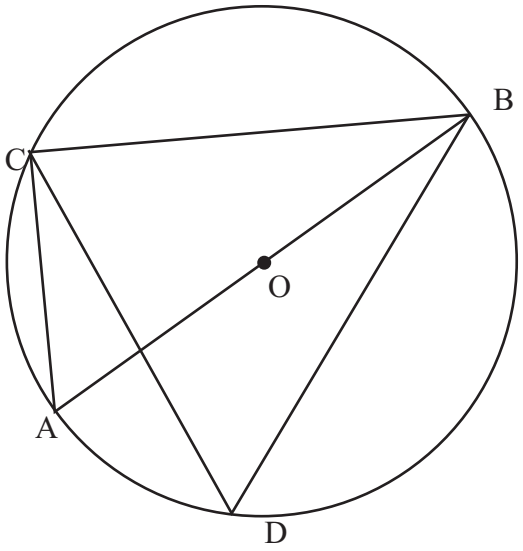
(i) Copy the diagram and mark the data.

(ii) Write the relationship between two line AB and CD. Write the reason.

(iii) Show that BCE $\Delta \equiv$ BED Δ.

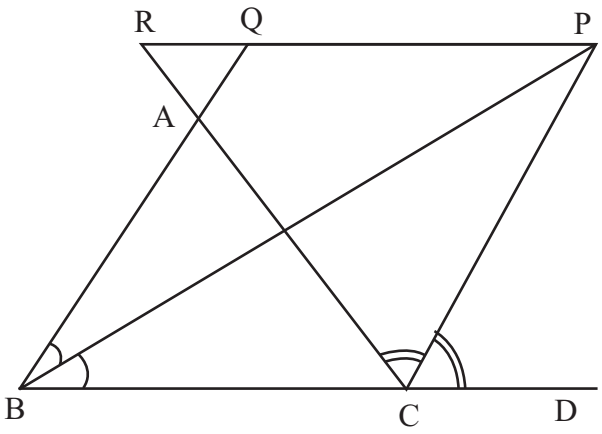
(iv) Show that the two triangles AEC and DEB are equiangular.

(v) If CD = 10cm, find the value of AE . BE.



(12) BC side of the triangle ABC has been produced up to D. Angular bisectos of ABC and ACD meets at P. The line drawn parallel to BC through P meet BA and CA at Q and R respectively. Copy the diagram in your answer.

Show that, QR = BQ - CR



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දෙවන වාර පරීක්ෂණය - 10 -11 ශ්‍රේණි - 2025

Second Term Test - Grade 10 -11 - 2025

ගණිතය II

විෂයය Subject	විභාග අංකය Index No.								

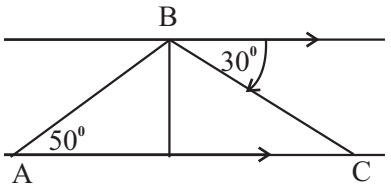
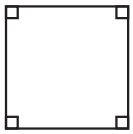
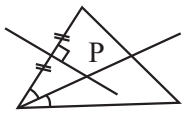
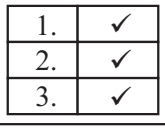
විභාග ශාලාවෙන් පිටතට ගෙනයාම තහනම්. Not to be removed from the Examination Hall.

Provincial Department of Education - NWP

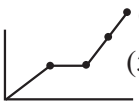
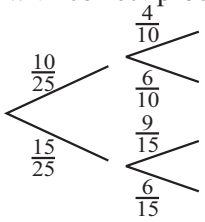
Grade 11 | **Second Term Test - 2025** | **Mathematics**

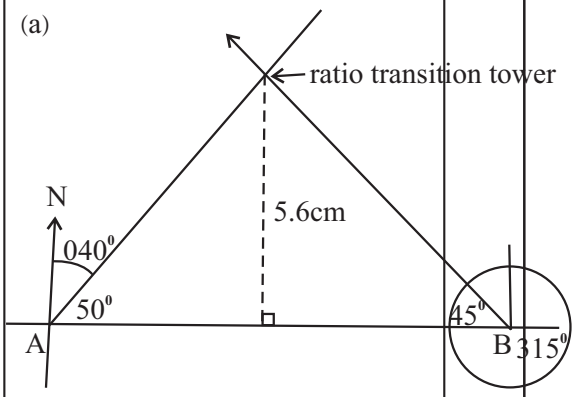
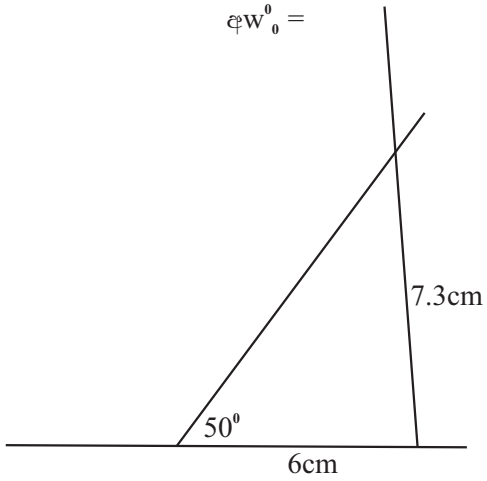
Answer sheet

01.	8.1		02
02.	$3x + 2x = 90^0$ $5x = 90^0$ $x = 18^0$ $(3x + 2x + 90^0 = 180^0)$ සඳහා ලකුණු 1 ක් දෙන්න.	01 01	 02
03.	$24a^2b^2$ $4a^2b^2 = 2 \times 2 \times a \times a \times b$ $6ab^2 = 2 \times 2 \times a \times b \times b$ $8ab = 2 \times 2 \times 2 \times a \times b$ සඳහා ලකුණු 1 ක් දෙන්න.		02
04.	AB = AC ගැනීමට S.A.S. එක්වරම ලියා ඇති විට	01	 02
05.	$2^{-3} = \frac{1}{8}$		02
06.	$\frac{2(x-2)}{2x} \times \frac{3}{(x-2)}$ 1 marks for writing common factor or reciprocal. $\frac{3}{x}$	01 01	 02
07.	30^0 වෙනත් ඕනෑම නිවැරදි ක්‍රමයකට ලකුණු දෙන්න.		02
08.	$\frac{12\,000}{30\,000} \times 100\%$ 40%	01 01	 02
09.	$\left(\frac{1}{2} \times 2 \times \frac{22}{7} \times 7\right) + 3 \times 14$ හෝ $= (22) + (42)$ = 64 cm	01 01	 02
10.	$m = \frac{6-4}{0-2}$ $m = -1$ ට හෝ $C = 6$ ට ල.01 $= \frac{2}{-2}$ $= -1$ $C = 6$ $y = -1x + 6$ හෝ $y = -x + 6$	01 01	 02
11.	QS = 4cm OR = 5cm QR = 6m ..... Triple. OS = 3cm Perimeter = 4 + 5 + 6 + 3 = 18 cm	01 01	 02
12.	$\frac{450}{150^0} \times 60^0$ 180	01 01	 02
13.	$\frac{180^0 + 40^0}{2} = \frac{220^0}{2} = 110^0$		02
14.	80 x 20 1600 *l*	01 01	 02

15.	$3x^2 - 9x + 2x - 6$ හෝ $3x(x - 3) + 2(x - 3)$ හෝ $(x - 3)\ 3x + 2$ එක්වරම $3x + 2$ ගෙන ඇත්නම් ලකුණු 2 දෙන්න.	01 01	02
16.	$T_{15} = ar^{14}$ $a = 3$ හෝ $= 3 \times 3^{14}$ $r = 3$ $= 3^{15}$	01 01	02
17.	 නිවැරදිව ලකුණු කිරීමට 50^0 30^0	 01 01	 02
18.	$P(A \cap B)' = \frac{5}{6}$		02
19.	 නිවැරදි 3 එකක් හෝ වැරදි නම් ලකුණු නොදෙන්න.		02
20.	 ශීර්ෂ කෝණ 90^0 ලෙස ලකුණු කළ මිනුම් සහිත දළ රූපයට පමණක් ලකුණු 02 දෙන්න.		02
21.	$S\hat{P}Q = 40^0$ ගැනීමට $P\hat{S}Q = 90^0$ ගැනීමට $90^0 - 40^0$ ගැනීමට ල.01 හෝ 50^0 Marking the angle and obtaining 50^0		02
22.	 Drawing perpendicular bisector of AB Marking P	01 01	02
23.	 නිවැරදිව ලකුණ දැමීම එකක් හෝ වැරදි නම් ලකුණු නොදෙන්න.		02
24.	volume $= \frac{1}{2} \times 12 \times 8 \times 10$ $= 480\ cm^3$	01 01	02
25.	$2 \times \frac{2}{2x} + \frac{3}{2x} = \frac{7}{4}$ getting equal denominator $\frac{4}{2x} + \frac{3}{2x} = \frac{7}{4}$ පිළිතුර 2 සඳහා $\frac{7}{2x} = \frac{7}{4}$ $2x = 4$ $x = 2$	01 01	02
			50

Answer sheet

	I පත්‍රය - B කොටස		
01.	(i) $\frac{5}{7}$		02
	(ii) $\frac{5}{7} \times \frac{3}{5} = \frac{3}{7}$ ගැනීමට	01	
	$\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$	01	
	$1 - \frac{5}{7} = \frac{2}{7}$	01	03
	(iii) $\frac{2}{7} \rightarrow 42\,000$	01	
	$\frac{3}{7} \rightarrow 63\,000$		01
	$\therefore \frac{1}{7} \rightarrow 21\,000$	01	
	$\therefore \frac{7}{7} \rightarrow 7 \times 21\,000$		02
	= රු. 147 000	01	
	(iv) $\frac{142\,000}{5} \times 100$ % (බෙදීමට ල.)	01	
	රු. 2940 000	01	02
			10
02.	(a) (i) 400	01	
	30	01	02
	 (60, 1400)	01	
	(50, 1000)	01	02
	(ii) මධ්‍යන වේගය = $\frac{1400}{60}$		
	= $23\frac{1}{3}$ ms^{-1}		01
	(b) (i) arc length = $\frac{1}{4} \times 2 \times \frac{22}{7} \times 28$		
	= 44 m	01	
	Perimeter = 44 + 88 + 28 + 88 + 28		
	= 276 m	01	03
	(ii) $\frac{1}{4} \times \frac{22}{7} \times 28 \times 28 + 88 \times 28$		
	= 616 + 2464	1+1	
	$\frac{3080}{616}$ = 5 times	01	03
			10
03.	(a) (i) 18 x 10 = 180 මී-දි.	01	
	(ii) 20 x 4 = 80	01	
	180 - 80 = 100 මී-දි.	01	
	(iii) No. of remaining days = 8 - 4		
	= 4	01	
	no. of men = $\frac{100}{4}$	01	
	= 25	01	
	25 - 20 = 5 men	01	07
	(b) $\frac{23\,200}{80\,000} \times 100\%$	1+1	
	29%	01	03
			10
04.	(a) (i) ජනක අංක 3 බෝලයත්, නීලන් අංක 4 බෝලයත් ගැනීමේ (බෝල දෙකෙහි අංකවල එකතුව 7 ලෙස ලියා ඇත්නම් ලකුණු නොදෙන්න.)	02	
	(ii) To complete the grid	03	05
	(b) (i) one mark for one branch with correct probability		
		03	
	(ii) $\frac{15}{25} \times \frac{9}{15} = \frac{135}{375}$	02	05
	හෝ $\frac{9}{25}$ ට		**10**
05.	(i) 6 සහ 17		02
	(ii) Correct cumulative frequency curve		
	marking points		02
	drawing curve		02
	(iii) obtaining median mark		02
	(iv) mark which is nearest and more than Q_3		02
			10
	II පත්‍රය - A කොටස		
01.	Compound interest for 1st year		
	= $\frac{12}{100} \times 100\,000$	01	
	= රු. 12 000	01	
	interest for 2nd year		
	= $\frac{12}{100} \times 112\,000$		
	= රු. 13 440	01	
	total interest = 12 000 + 13 440		
	= Rs. 25 440	01	

	no. of shares = $\frac{100\,000}{25}$ = 4 000	01	
	dividend for 1st year = 4.50 x 4000 = රු. 18 000	01	
	dividend for 2nd year = 4 000 x 5 = රු. 20 000	01	
	Amount received by selling shares = 27 x 4 000 = රු. 108 000	01	
	total profit = 18 000 + 20 000 + 8 000 = රු. 46 000	01	
	pure profit = 46 000 - 25 440 = රු. 20 560 20 500 < 20 560	01	**10**
02.	(i) 1		01
	(ii) marking axes	01	
	mark at least 6 point	01	
	smooth curve	01	03
	(iii) (a) (0.2, 0), (3.7, 0)	1+1	02
	(b) $y = (x - 2)^2 - 3$ P = 2 ද Q = 3 ගේ.	01	01
	(iv) (a) -0.6 < x < 4.6	1+1	02
	(b) AB = 4.6 - (-0.6) = 4.6 + 0.6 = 5.2 AB ⩾ 5.2	01	**10**
03.	(a) 		
	scale length 10 cm	01	
	040⁰	01	
	315⁰	01	
	P පිහිටීම	01	
	36 = 5.6 + 10 = 56 m (±0.1cm)	01	05
	(b) ඇW⁰₀ = 		
	(b) scale length 6cm	01	
	50⁰	01	
	Drawing perpendicuiar	01	
	scale height 7.3cm	01	
	(± 0.1cm)		
	7.3 x 5 = 36.5m	01	05
			10
04	(i) $9^2 - \frac{1}{2} \times x \times (x+2) = 60$	01	
	$81 - \frac{1}{2} x (x+2) = 60$		
	$162 - x^2 - 2x = 120$	01	02
	$x^2 + 2x + 120 - 162 = 0$		
	$x^2 + 2x - 42 = 0$		
	$a = 1, b = 2, c = -42$		
	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	01	
	$= \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times 1 (-42}}{2 \times 1}$	01	
	$= \frac{-2 \pm \sqrt{4 + 168}}{2}$		
	$= \frac{-2 \pm \sqrt{172}}{2}$	01	
	$= \frac{-2 \pm 2\sqrt{43}}{2}$	01	
	$= \frac{-\cancel{2}(-1 \pm \sqrt{43})}{\cancel{2}}$		
	$= -1 \pm 6.56$	01	
	$x = -1 + 6.56$ = 5.56 cm, $x > 0$	$x = -1 - 6.56$ = - 7.56, $x < 0$	
	$x = 5.6$ cm	01	06

සියලුම විෂයන්ට අදාළ ප්‍රශ්න පත්‍ර/පිළිතුරු පත්‍ර/සටහන් ලබා ගැනීමට Past Papers Wiki වෙත පිවිසෙන්න

	(b) length of hypotenuse $=\sqrt{x^2 + (x+2)^2}$ $5.6^2 = 31.36$ $=\sqrt{5.6^2 + 7.8^2}$ $7.8^2 = \frac{60.84}{92.20}$ $= 9.6$ cm $9 < 9.6$	01 01	02
			10
05	$x + y = 150$ $250x + 300y = 40\ 800$ $x + y = 150$ —① (solving by removing one variable) $5x + 6y = 316$ —② ① + 6 $6x + 6y = 900$ —③ ③ - ② $x = 84$ $84 + y = 150$ $y = 150 - 84$ $= 66$	02 02 03 02 01	
			10
06	(i) 500 - 600 (ii)	01	01

ප.පු.	මැ.	d	f	f x d	
200-300	250	-300	1	-300	
300-400	350	-200	2	-400	-1100
400-500	450	-100	4	-400	
500-600	550	0	8	0	
600-700	650	+100	6	+600	
700-800	750	+200	4	+800	+3100
800-900	850	+300	3	+900	
900-1000	950	+400	2	+800	
			30	$\Sigma fd = 3100 - 1100 = 2000$	

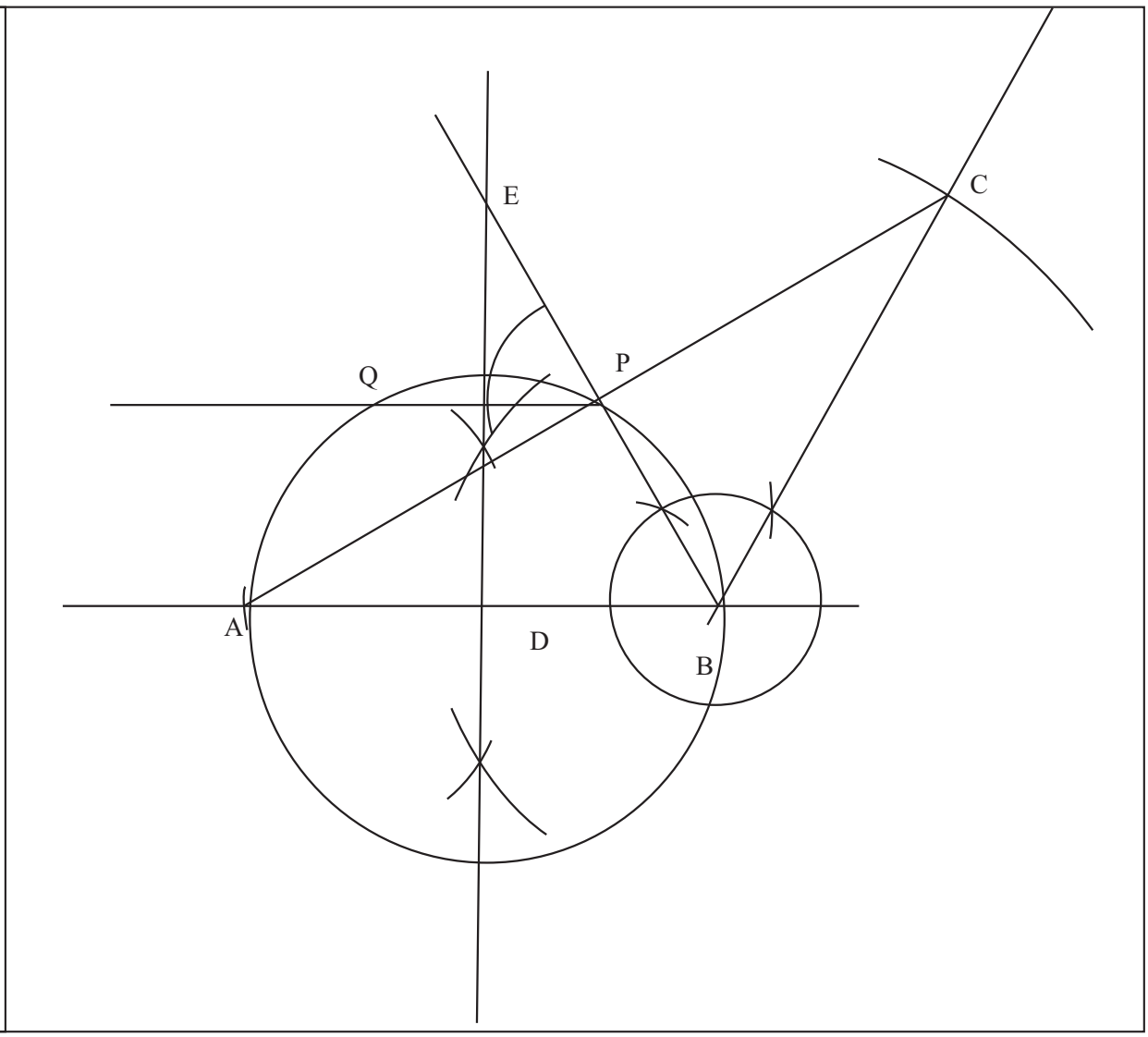
mean $= 550 + \frac{2000}{30}$	01	
$= 550 + 66.6$		
$= 616.6$		
Ω 617		
mean value	01	
d	01	
fd	01	
Σfd	01	
ආසන්න අගයට	01	05

(වැරදි 2 ක් නොසලකන්න)

	(iii) 5 x 617 x 90 = රු. 277 650	01 01	02
	(iv) 300x1 = 300 400x2 = 800 500x4 = 2000 600x8 = 4800 700x6 = 4200 800x4 = 3200 900x3 = 2700 1000x2 = 2000 15280 15000 < 15280	 01 01	 02
			10
	II පත්‍රය - B කොටස		
07	(i) 5, 13, 21, ... $a = 5, d = 8, n = 7$ $T_n = a + (n - 1) d$ $T_n = a + 6d$ $= 5 + 6 \times 8$ $= 53$	 01 01 01	 03
	(ii) $93 = 5 + (n - 1) + 8$ $93 = 5 + 8n - 8$ $93 = 8n - 3$ $8n = 93 + 3$ $8n = 96$ $n = 12$	 01 01	 02
	(iii) $T_{51} = 5 + 8 \times 50$ $= 5 + 400$ $= 450$ 10, 20, 30, ... $450 = 10 + (n - 1) + 10$ $450 = 10 + 10n - 10$ $n = 45$ කණ්ඩ 45 + 1 = 46 හමුවේ.	01 01 01 01 01	 02 03
			10

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08



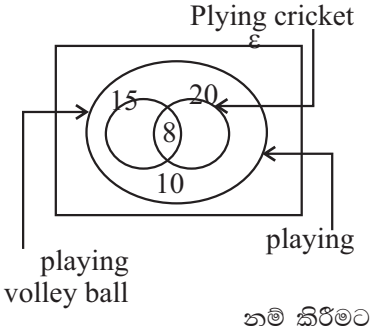
	(i) AB	01	
	120^0	01	
	BC	01	03
	(ii) perpendicular bisector	01	
	D	01	02
	(iii) angular bisector	01	
	E/P	01	02
	(iv) semicircle	01	01
	(v) parallel line	01	01
	(vi) APQ = 30^0	01	01
			10
09	$\frac{1}{3}$ x a^2 x $3a$ x 12 = $\frac{4}{3}$ πr^3	1+1	02
	$a^3 = \frac{\pi}{9}$ r^3		

	$a^3 = \frac{3.14}{9}$ x 2.84^3		
	$lg\ a^3 = lg\ 3.14 + 3\ lg\ 2.84 - lg\ 9$	1	
	$lg\ a^3 = 0.4914 + 3$ x $0.4533 - 0.9542$	1+1+1	
	$lg\ a^3 = 0.4914 + 1.3599 - 0.9542$		
	$lg\ a^3 = 1.8513 - 0.9542$	1	
	$lg\ a^3 = 0.8971$	1	
	a^3 = ප්‍රතිලඝු (0.8971)		
	$a^3 = 7.89$	1	
	a^3 Ω 8 cm		
	a = 2 cm	1	08
			10

සියලුම විෂයන්ට අදාල ප්‍රශ්න පත්‍ර/පිළිතුරු පත්‍ර/සටහන් ලබා ගැනීමට Past Papers Wiki වෙත පිවිසෙන්න

Past Papers Wiki

10 (i)

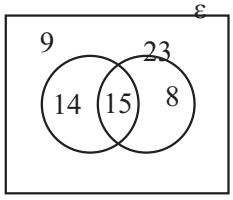


නම් කිරීමට 02
20 01
15 01
10 01 05

(ii) 12 01 01

(iii) 37 + 9 = 46 01 01

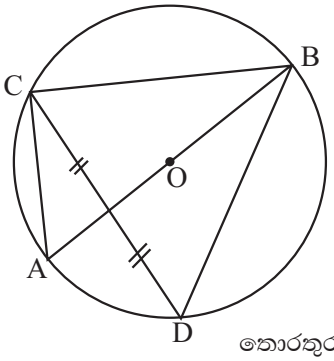
(iv)



14 01
9 01
29
46 01 03

10

11 (i)



තොරතුරු 01

(ii) $CD \perp AB$ 01
වචනයෙන් ලියා තිබුනත් ලකුණු දෙන්න.

(iii) BCE Δ හා BED Δ ගත් විට
CE = ED (data) 01
$B\hat{E}C = B\hat{E}D = 90^0$ 01
(OE ⊥ CD බැවින්)
BO = BO (common side)
∴ BEC Δ ≡ BED Δ ට
(S. A. S. අවස්ථාව) 01 03

(iv) AEC Δ හා DEB Δ showing equiangular
$A\hat{C}E = E\hat{B}D$ (angle in same segment)
$A\hat{E}C = B\hat{E}D$ (vertically opposite ∢) 01
$C\hat{A}E = E\hat{D}B$ (angle in same segment 01) 02
∴ AEC Δ හා DEB Δ are equiansular.

(v) $\frac{AE}{ED} = \frac{AC}{BD} = \frac{CE}{BE}$ 01

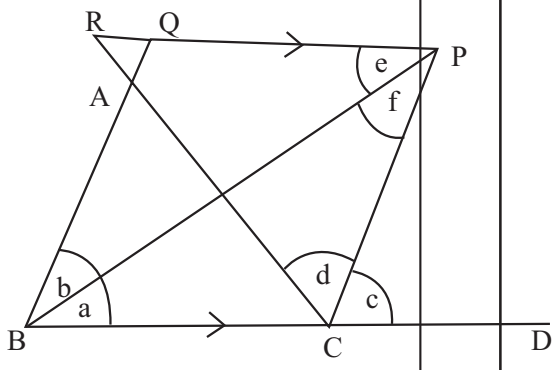
CD = 2 CE = 2 ED
CE = ED = 5 cm 01

$\boxed{\frac{AE}{5} = \frac{AC}{BD} = \frac{5}{BE}}$

AE . BE = 5 x 5 = 25 01 03

10

12



01

වීජ ගණිතය යොදා ගෙන සාධනය කරමු.

සාධනය :-

a = b (data) 01

නමුත් a = e (BC // QP alternate ∢) 01

∴ b = e (axioms)

∴ BQ = PQ (සමාන කෝණවලට 01
සම්මුඛ පාද සමානවේ.)

එලෙසම C = e + f (alternate ∢) 01

නමුත් C = d (data) 01

∴ d = e + f (axioms)

∴ PR = CR (∢වලට සම්මුඛ
පාද = වේ) 01

PR = PQ + QR

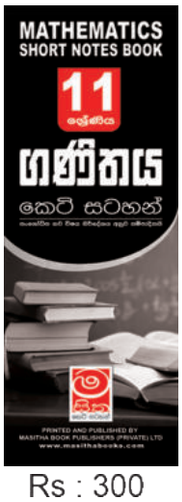
∴ PQ + QR = CR 01

BQ + QR = CR (PQ = BQ) 01

∴ QR = BQ - CR 01

10

11 ශ්‍රේණිය කෙටි සටහන් පොත් සංශෝධිත නව විෂය නිර්දේශයේ පාඩමෙන් පාඩමට



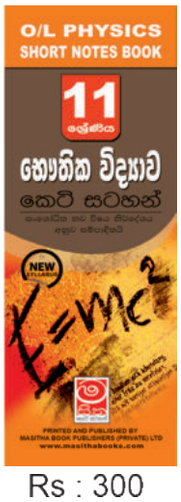
Rs : 300



Rs : 300



Rs : 300



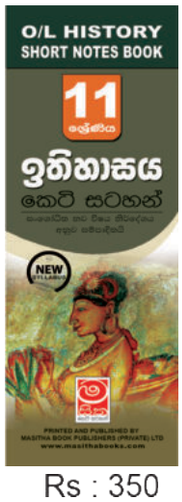
Rs : 300



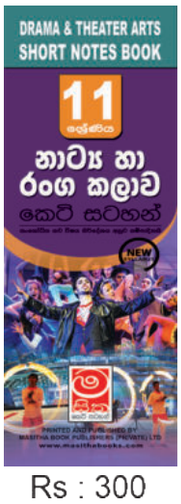
Rs : 300



Rs : 300



Rs : 350



Rs : 300



Rs : 300



Rs : 200



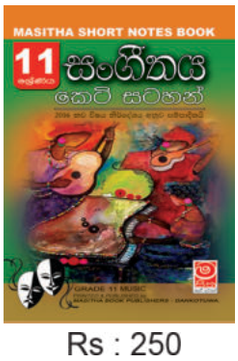
Rs : 200



Rs : 200



Rs : 250



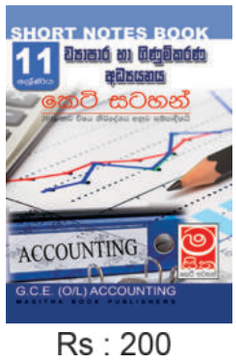
Rs : 250



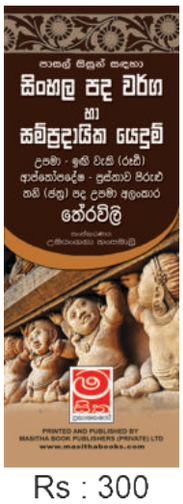
Rs : 200



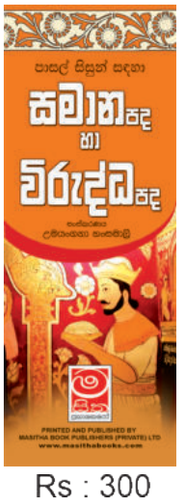
Rs : 200



Rs : 200



Rs : 300



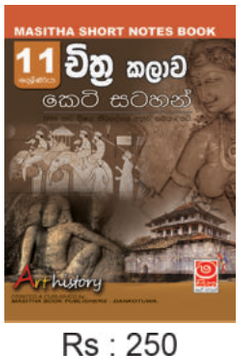
Rs : 300



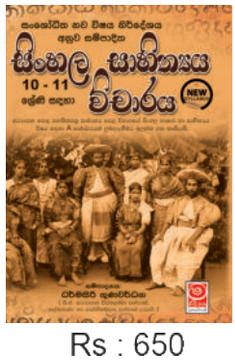
Rs : 200



Rs : 200



Rs : 250

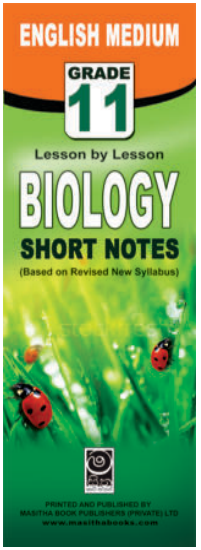


Rs : 650

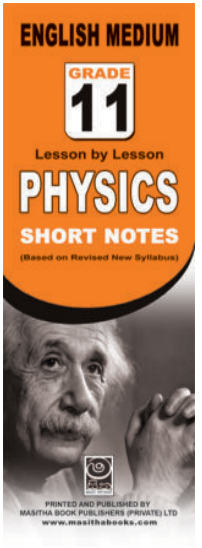
මෙම කෙටි සටහන් පොත් සකස් කොට ඇත්තේ ජාතික අධ්‍යාපන ආයතනය මගින් සම්පාදනය කොට ඇති නව විෂය නිර්දේශය අනුව, විෂය ප්‍රවීණත්වයෙන් යුත් සම්පාදක මණ්ඩලයක් විසිනි, එහෙයින් මෙහි අන්තර්ගත සියලු සටහන් එම විෂය නිර්දේශයේ සෑම මාතෘකාවක්ම ආවරණයකොට තිබේ. මෙම ග්‍රන්ථ සම්පාදනය කිරීමේ දී පූර්ණ අවධානය යොමුකොට ඇත්තේ සිසුන්ට මෙන්ම ගුරුවරුන්ට ද අවශ්‍යයෙන්ම මතකයේ තබා ගත යුතු සියලු කරුණු අන්තර්ගත කරමින් වන හෙයින් වෙනත් අතිරේක ග්‍රන්ථ පරිහරණයකින් තොරව වුවද අදාල විෂය නිර්දේශයේ වැදගත් කරුණු සියල්ල ඉතා පහසුවෙන් හැදෑරීමේ මෙන්ම මතකයේ රඳවා ගැනීමේ හැකියාව ද ලැබෙනු ඇත.

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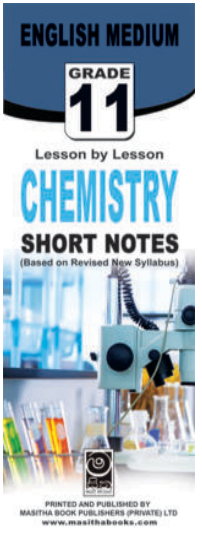
11 ශ්‍රේණි ඉංග්‍රීසි මාධ්‍ය කෙටි සටහන් පොත් ENGLISH MEDIAM SHORT NOTE BOOKS Rs 180/-



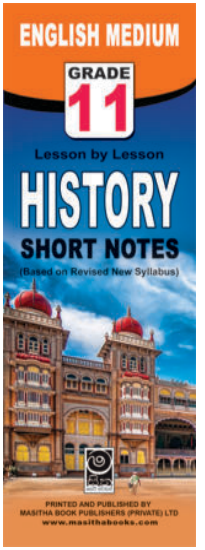
Rs : 180



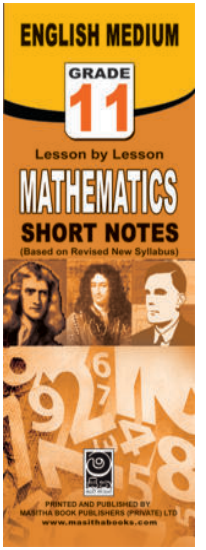
Rs : 180



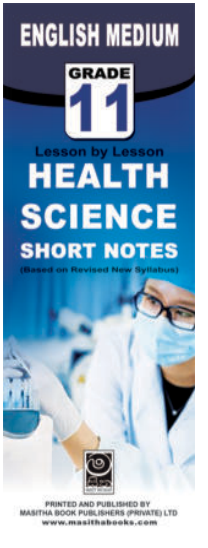
Rs : 180



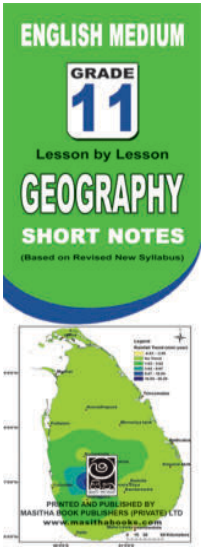
Rs : 180



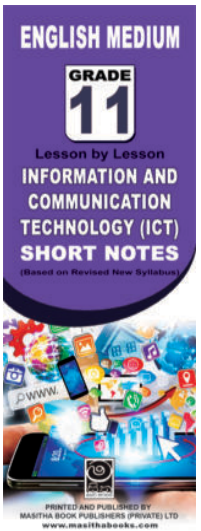
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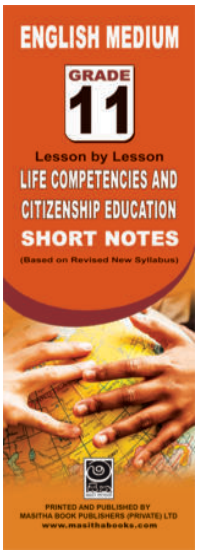
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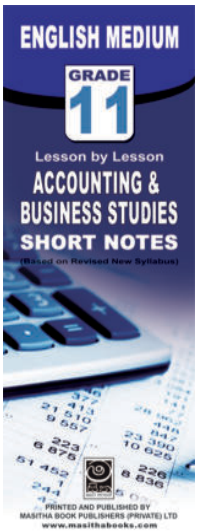
Rs : 180



Rs : 180



Rs : 180



Rs : 180

මෙම කෙටි සටහන් පොත් සකස් කොට ඇත්තේ ජාතික අධ්‍යාපන ආයතනය මගින් සම්පාදනය කොට ඇති නව විෂය නිර්දේශය අනුව, විෂය ප්‍රවීණත්වයෙන් යුත් සම්පාදක මණ්ඩලයක් විසිනි, එහෙයින් මෙහි අන්තර්ගත සියලු සටහන් එම විෂය නිර්දේශයේ සෑම මාතෘකාවක්ම ආවරණයකොට තිබේ. මෙම ග්‍රන්ථ සම්පාදනය කිරීමේ දී පූර්ණ අවධානය යොමුකොට ඇත්තේ සිසුන්ට මෙන්ම ගුරුවරුන්ට ද අවශ්‍යයෙන්ම මතකයේ තබා ගත යුතු සියලු කරුණු අන්තර්ගත කරමින් වන හෙයින් වෙනත් අතිරේක ග්‍රන්ථ පරිහරණයකින් තොරව වුවද අදාල විෂය නිර්දේශයේ වැදගත් කරුණු සියල්ල ඉතා පහසුවෙන් හැදෑරීමේ මෙන්ම මතකයේ රඳවා ගැනීමේ හැකියාව ද ලැබෙනු ඇත.

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